

8. Cardiff Bay is a 2.0km² freshwater lake. The Bay is not tidal but visiting yachts, from all over Europe, can enter the Bay from the sea, through locks. The Bay is fed by two rivers, the Taff and the Ely.



The following public information notice has been posted in Cardiff Bay.

Killer shrimps

Routine environmental monitoring identified that an invasive species of shrimp, *dikerogammarus villosus*, or killer shrimp has established itself within Cardiff Bay. It has been found in several sites across the UK.

There is no risk to public health or pets, however this non - native species poses a threat to the aquatic wildlife in our rivers, streams and lakes.

Controls are in place to try to prevent the spread to other water bodies.

Water users are advised to 'Check, Clean and Dry' all equipment and clothing, as they will not survive in dry conditions for more than 48 hours.



Examiner
only

- (a) (i) Invasive species are also known as alien species. Explain what is meant by the term *invasive species* and why they are a problem. [2]

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- (ii) In the public information notice opposite there is an error in the way in which the scientific name of the killer shrimp has been written. Identify the error by writing the scientific name in the correct form. [1]

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- (iii) Killer shrimps feed on native freshwater shrimps, aquatic insects and small fish. State how the biodiversity of the UK will be affected by the killer shrimp. [1]

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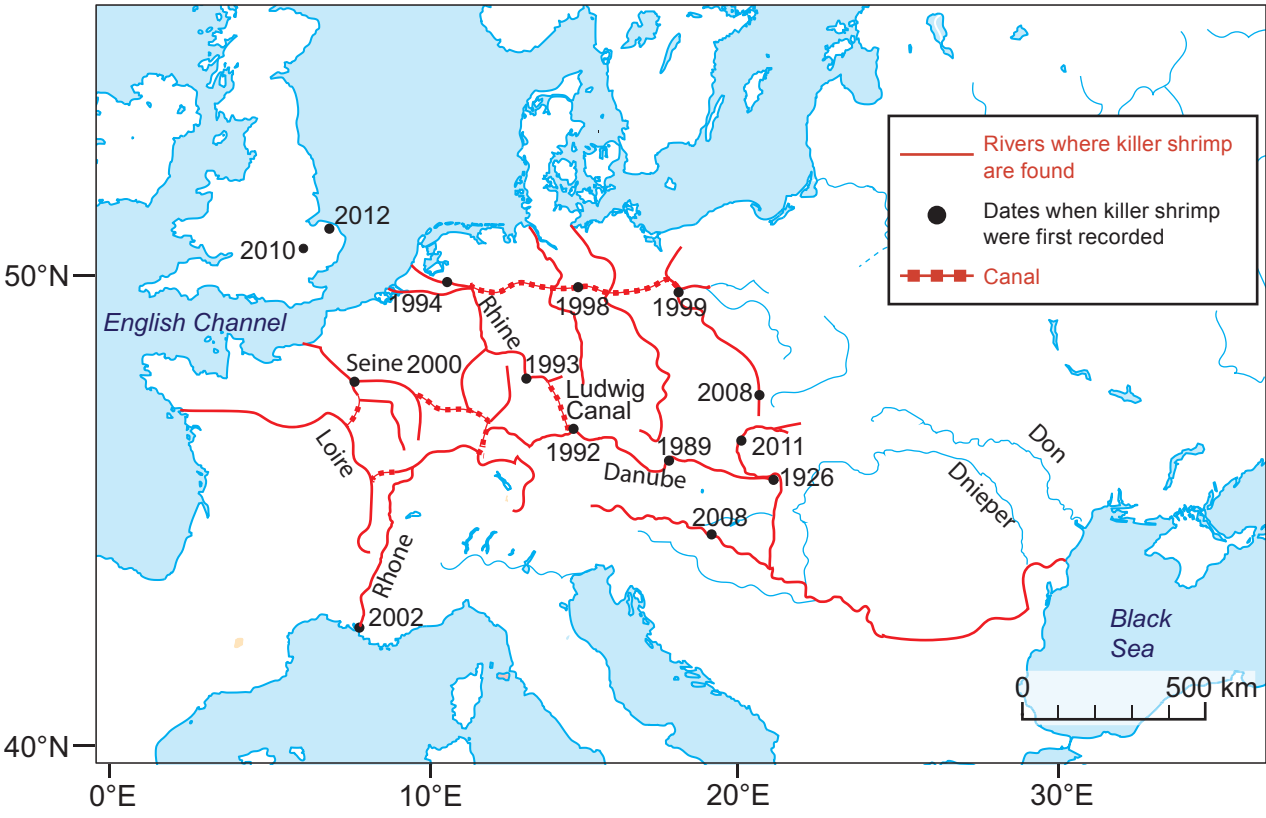
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(b) Killer shrimp fact file:

Environment	Freshwater reservoirs and lakes in UK. Cannot live in water with a salt content > 2.5 %
Native range	Black Sea, Russian Far East
Invasion pathway into UK	Unknown
Feeding	Predator
Status in UK	Established
Date of first record in UK	2010
Speed of spread	Downstream 124 km/year Upstream 35 km/year
Predators	Trout, perch, aquatic birds
Body length of adult	Up to 30 mm
Reproduction	Females lay about 50 eggs which hatch into young about 1.8 mm in length

The map below shows the spread of the killer shrimp westwards along the rivers and canals of Europe. It also shows the dates at which the killer shrimp was first recorded at various points.



Examiner
only

- (i) Use the map to suggest the route by which the killer shrimp travelled from the River Danube to the River Rhine and state the date when this occurred. [1]

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- (ii) The salt content of the English Channel is 3.4–3.5 %. State why the killer shrimp could not have travelled to Great Britain through the waters of the English Channel. [1]

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- (iii) As a trainee research biologist working for Natural Resources Wales you have been asked to study all the information in detail. Suggest **one** way by which the killer shrimp could have arrived in Great Britain. [1]

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- (iv) The River Taff rises in the Brecon Beacons and flows through Merthyr Tydfil to Cardiff Bay. Travelling along the River Taff, Merthyr Tydfil is 55.1 km from Cardiff Bay. If the killer shrimp were to travel up the River Taff from Cardiff Bay, calculate, to **two** significant figures, how long it would take to reach Merthyr Tydfil. [2]

time taken to reach Merthyr Tydfil = years

END OF PAPER

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